

Direct vs exchange THC block-encoding Toffoli comparison

Parameters: $N_{\text{up}} = 4$, $N_{\text{down}} = 22$, N_{IP} (central bond) = 256
 $N_k = k^3$ for k in $[1, 2, 3, 4, 5, 6]$ (so N_k in $[1, 8, 27, 64, 125, 216]$)
phase_bitsize $b = 32$

Two block encodings of the periodic THC tensor (arXiv:2601.16379 Eq. 9):

Exchange: bra-pair X reflections + central SVD block encoding of W^q .

Direct: orientation flipped -- inputs $(\mu, k_\mu), (\lambda, k_\lambda)$, outputs the other two. $|Q\rangle$ uniform LCU, two X reflections, two momentum modular adds $k \pm Q$, central DIAGONAL kernel $\sum W^Q_{\{IJ\}} |QIJ\rangle\langle QIJ|$ (QROAM $\rightarrow$ Ry $\rightarrow$ QROAM †), then the two $X^\dagger$.

Tuning policies (per curve):

T-opt: every QROAM `log_block_sizes` at the closed-form Toffoli optimum.

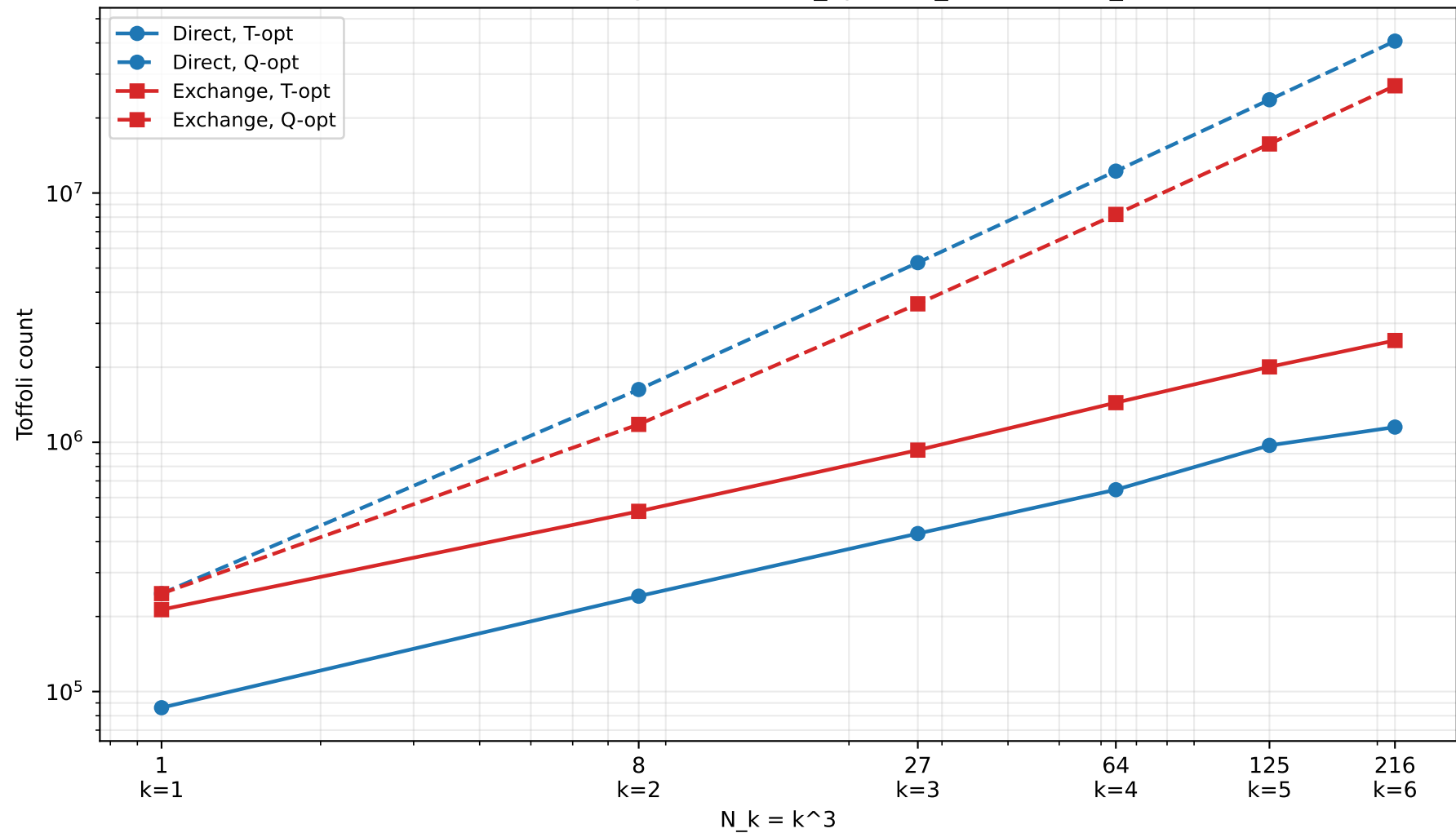
Q-opt: no QROAM blocking (`log_block_sizes = 0`) $\rightarrow$ minimal qubits.

All Toffoli / qubit counts come from Qualtran's `QECGatesCost / QubitCount` walking each bloq's `build_call_graph` (no analytic formulas).

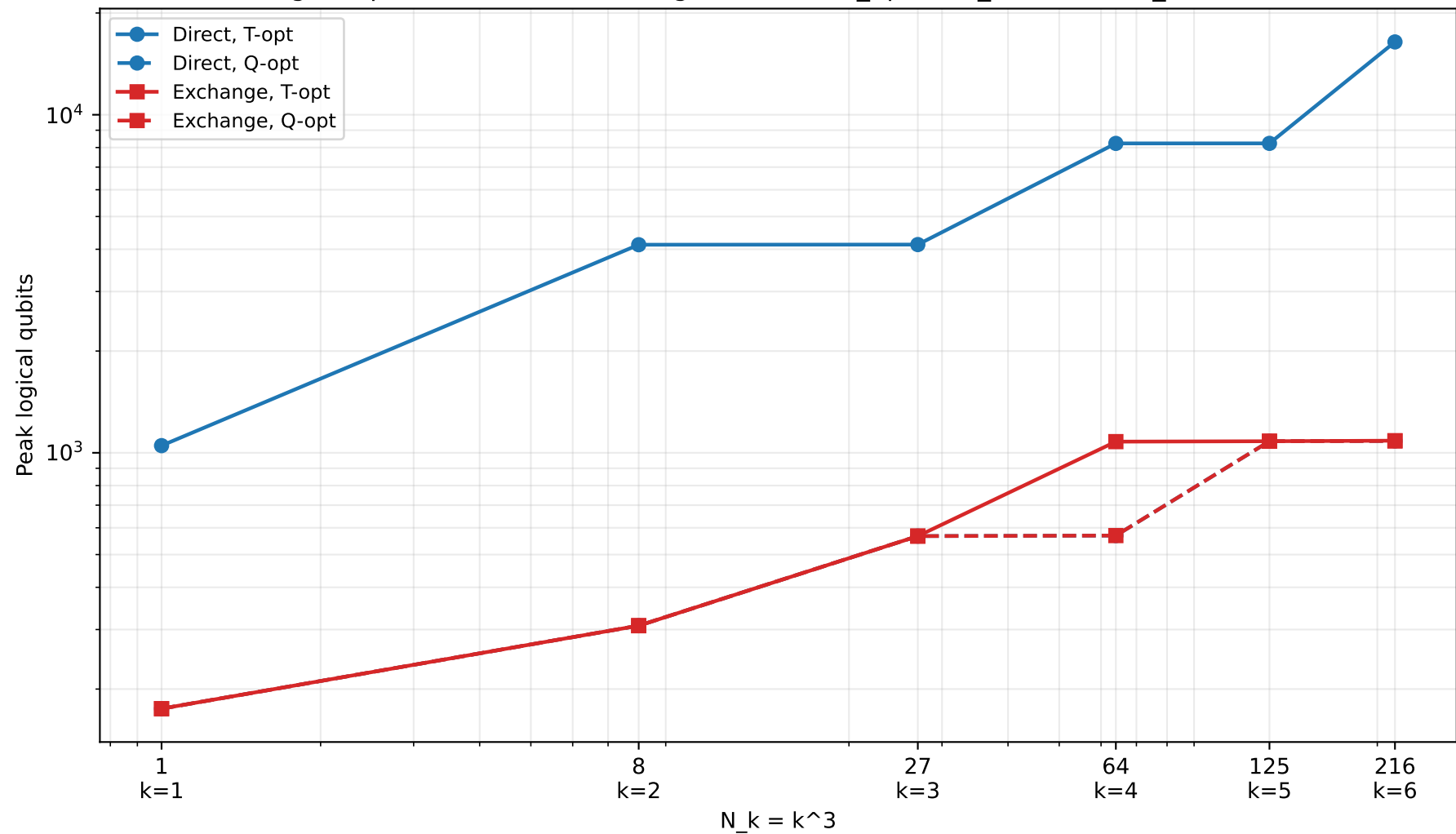
Note: the direct kernel's T-opt blocks a large $(N_k, N_{\text{IP}}, N_{\text{IP}})$ QROAM, so its

T-opt qubit count is much higher than its Q-opt one (a genuine $T \leftrightarrow Q$ tradeoff).

Toffoli count: direct vs exchange THC BE ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Peak logical qubits: direct vs exchange THC BE ($N_{up}=4$, $N_{down}=22$, $N_{IP}=256$, $b=32$)



Resource counts: direct vs exchange THC block encoding

k	N_k	Direct T-opt T	Direct T-opt Q	Direct Q-opt T	Direct Q-opt Q	Exch T-opt T	Exch T-opt Q	Exch Q-opt T	Exch Q-opt Q
1	1	86,130	1050	247,035	175	213,032	175	247,085	175
2	8	241,310	4126	1,627,655	308	528,414	308	1,179,709	308
3	27	430,334	4130	5,255,399	567	930,190	567	3,591,457	567
4	64	645,164	8227	12,239,117	569	1,441,878	1079	8,207,177	569
5	125	971,550	8229	23,668,215	1082	2,006,410	1082	15,732,277	1082
6	216	1,150,724	16422	40,680,941	1084	2,560,306	1086	26,921,005	1084